

# LIM-ARC: gent MoE

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gent OS

<https://slim.nexa-lang.com/>

# 0 About us

➤ gent OS (T2026105589911358)



➤ Proj59

➤ <https://slim.nexa-lang.com/>



# Catalogue

- ❖ 1 Background & Motivation
- ❖ 2 Project Introduction
- ❖ 3 SLIM-ARC Core Design
- ❖ 4 Projrct Implementation
- ❖ 5 Testing & Evaluation
- ❖ 6 Summary & Outlooking

1

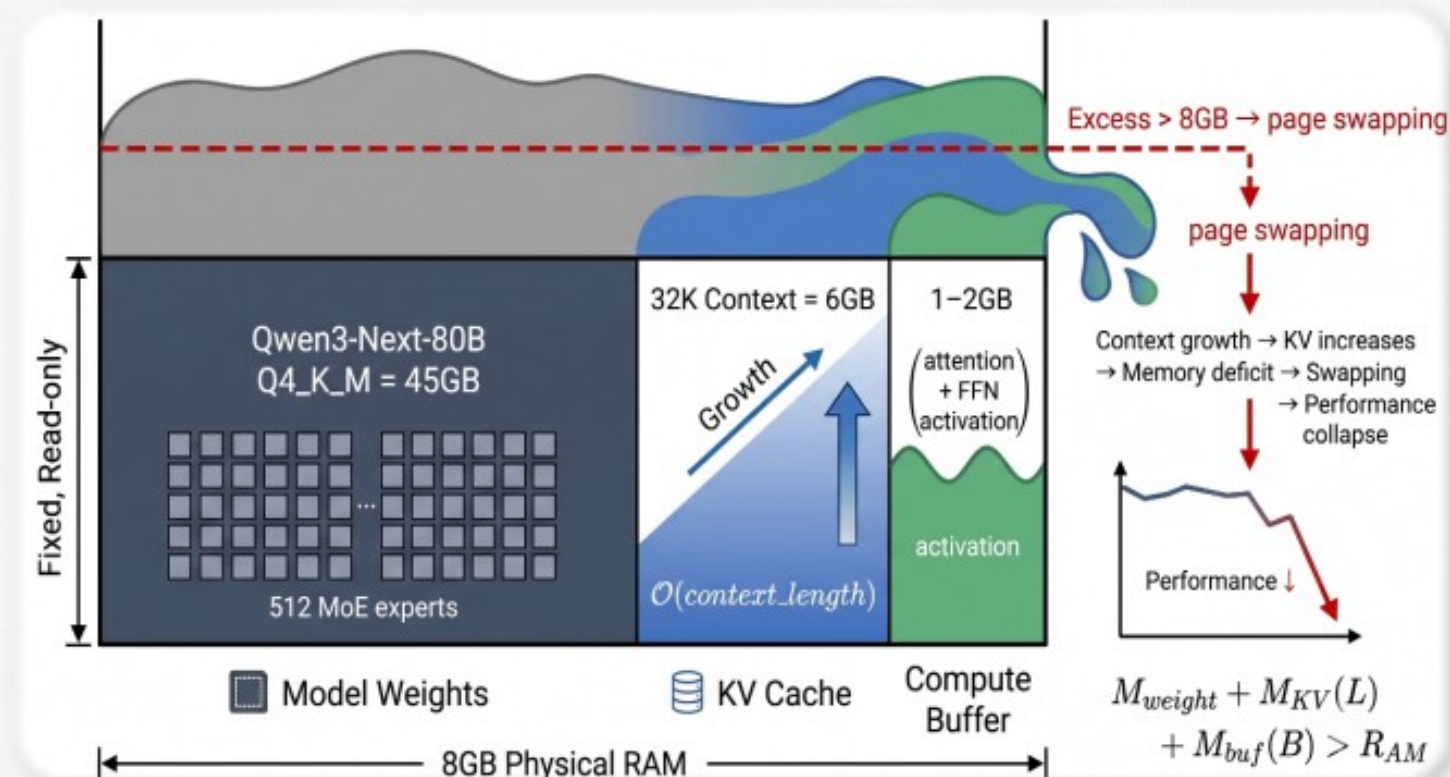
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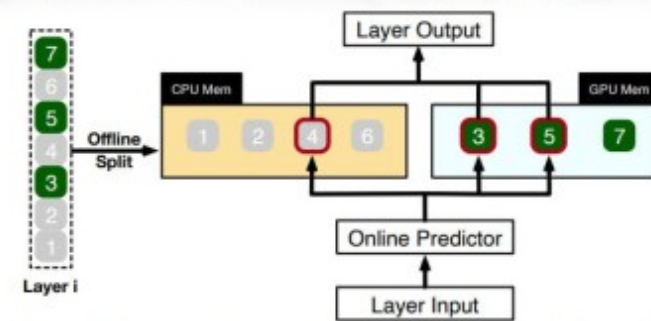
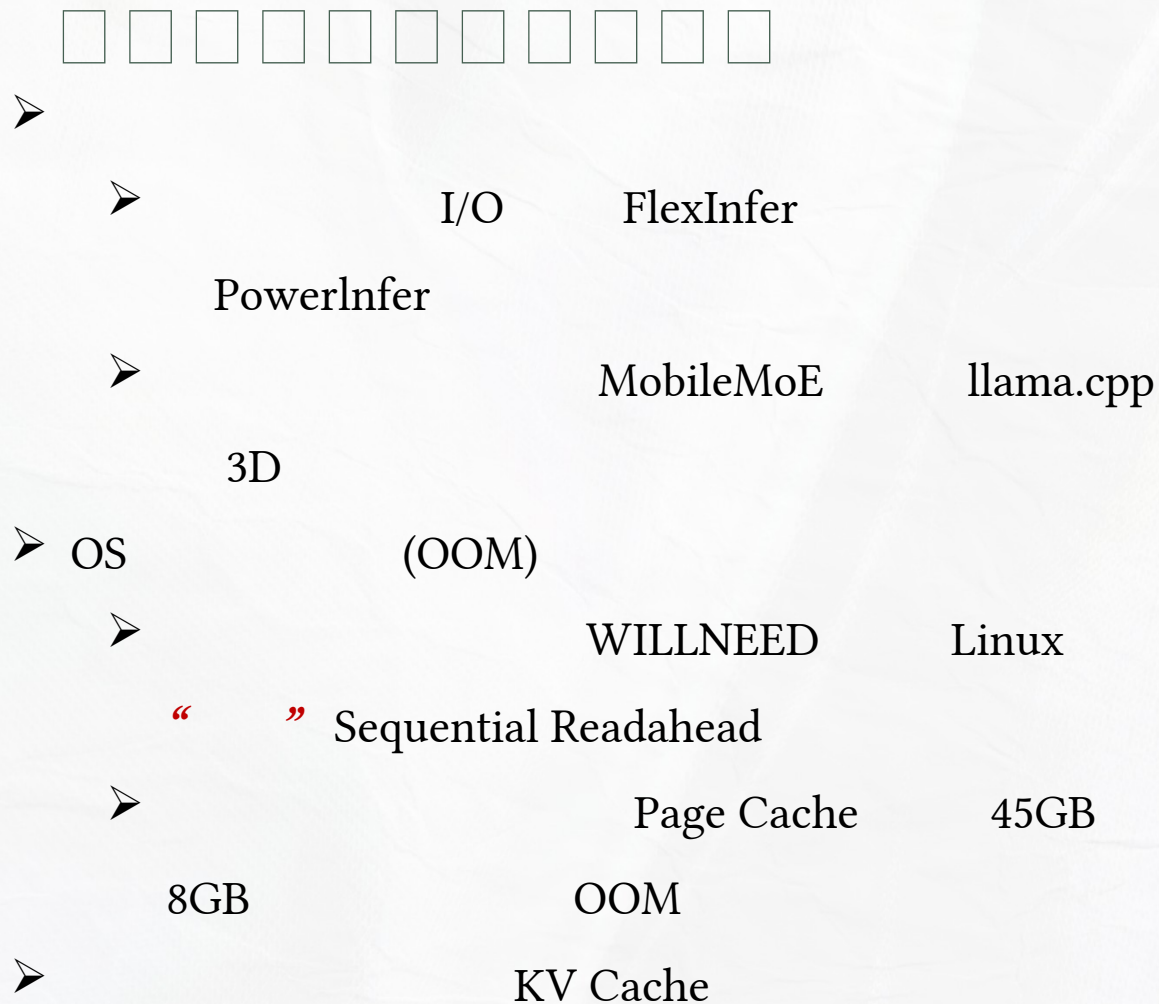
gent



80B Qwen3-  
Next-80B 45GB  
140GB  
8 16GB

KV Cache





**Figure 8:** An illustrative example shows how PowerInfer calculates different neurons for one LLM layer.

## 4.2 Single Layer Example

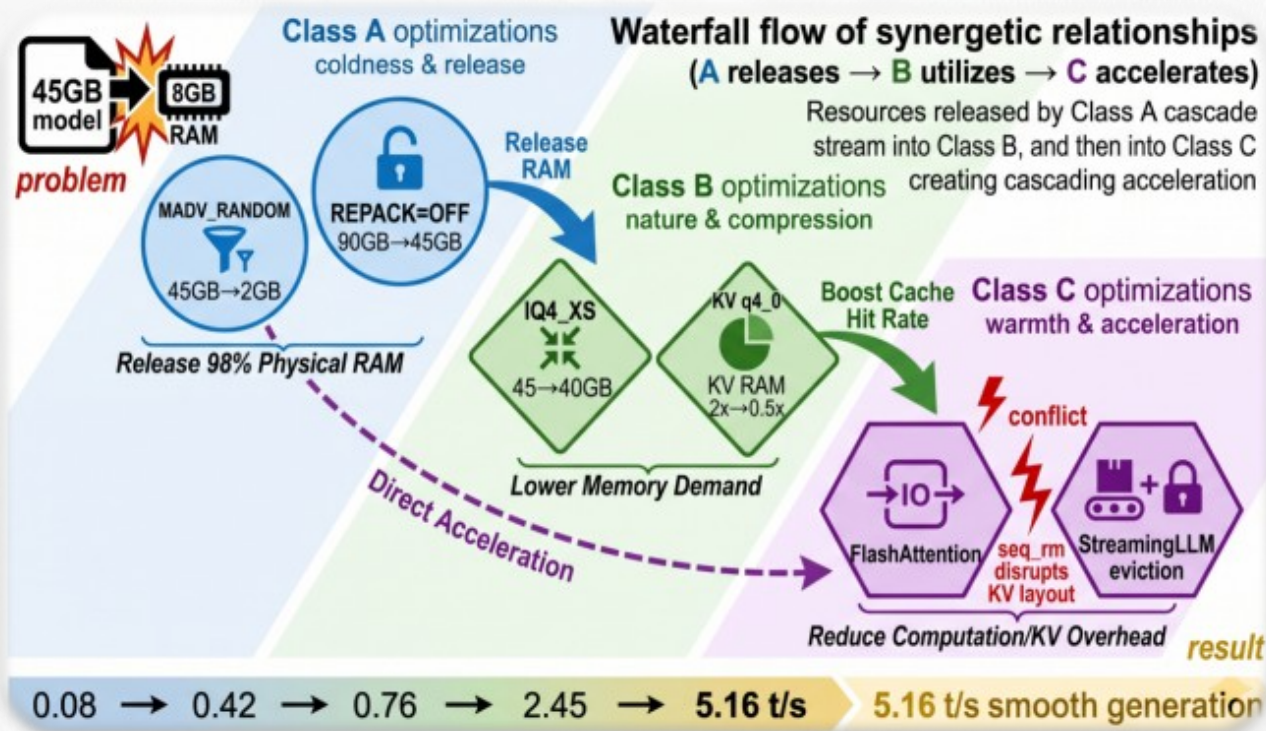
Figure 8 illustrates how PowerInfer coordinates GPU and CPU in processing a layer’s neurons. It classifies neurons based on offline data, assigning hot-activated ones (e.g., indices 3, 5, 7) to GPU memory and others to CPU memory. Upon receiving an input, a predictor identifies which neurons in the current layer are likely to be activated. For instance, it predicts activation for neurons 3, 4, and 5. It is crucial to note that hot-activated neurons, identified through offline statistical analysis, may not consistently match the runtime activation behaviors. For example, neuron 7, though labeled as hot-activated, is forecasted to be inactive in this case.

Both CPU and GPU then process predicted active neurons, ignoring inactive ones. The GPU computes neurons 3 and 5, while the CPU handles neuron 4. Once neuron 4’s computation is complete, its output is sent to the GPU for result integration.

2

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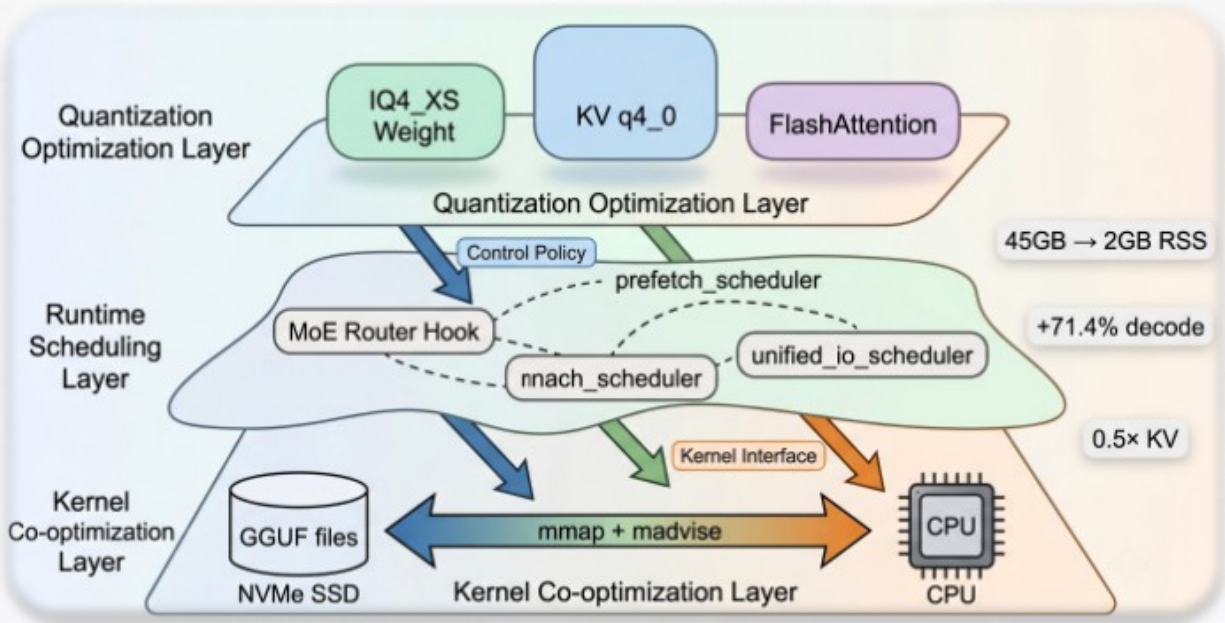
- OS
- (Cascading Synergy)
- (Class A) OS
- (Class B) RAM
- (Class C) I/O
- “ → → ”
- MoE



3

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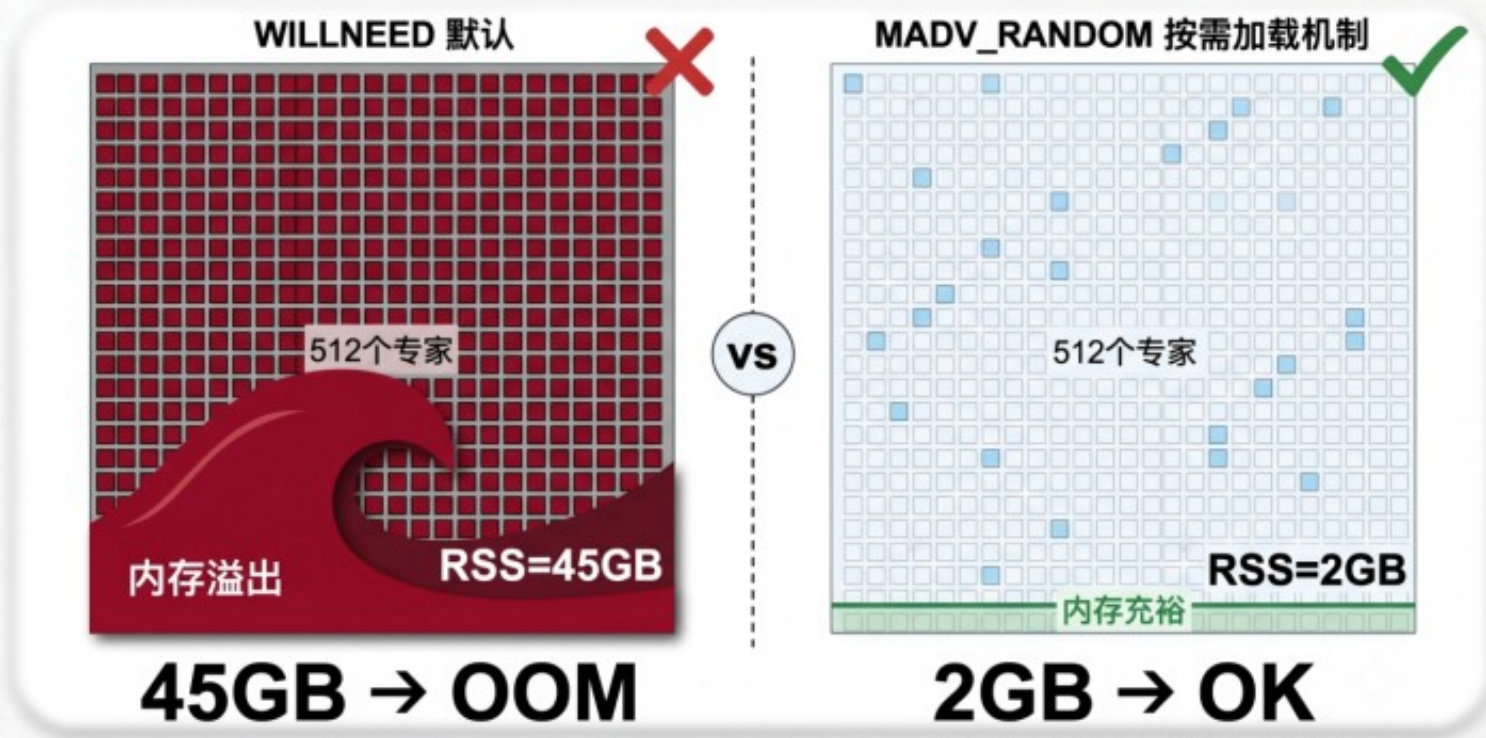
SLIM- RC

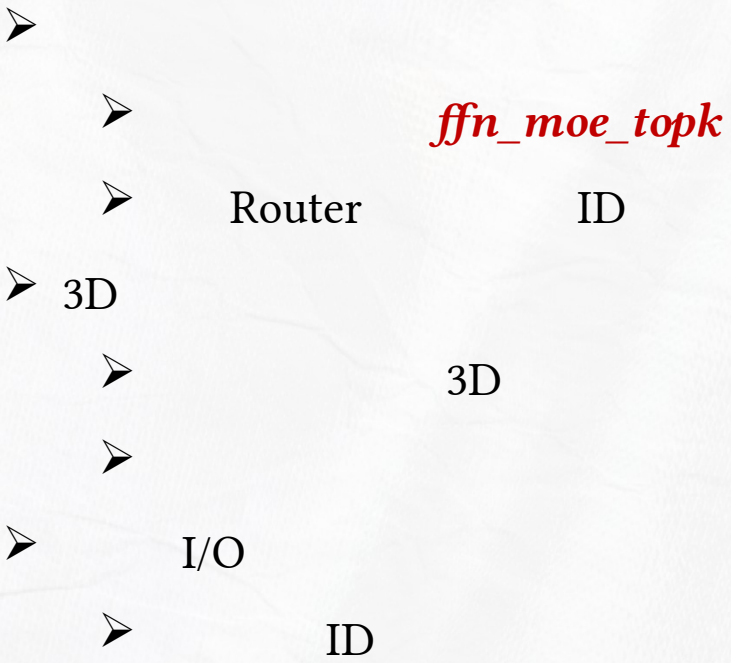
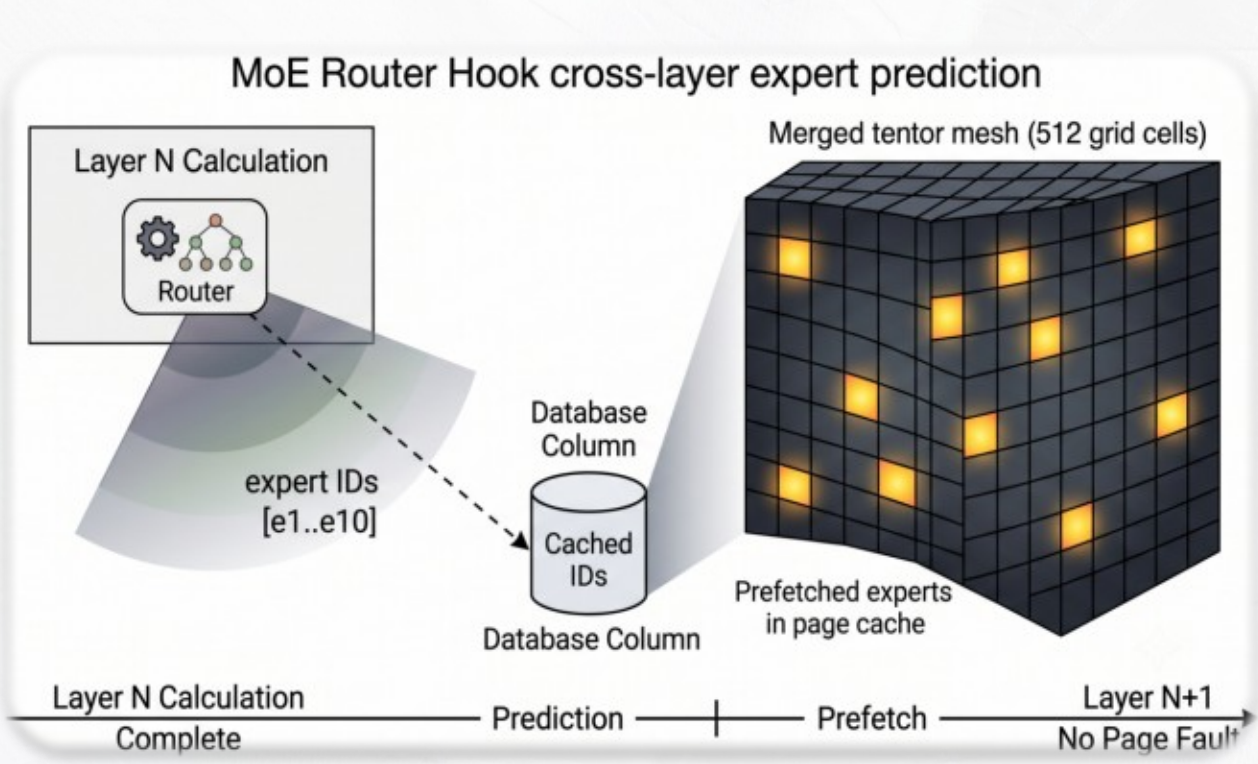


- OS
- 
- 
- 
- ***MMAP*** 45GB
- ***POSIX***
- MoE KV
- I/O
- IQ4\_XS ***FlashAttention***

# MADV\_RANDOM

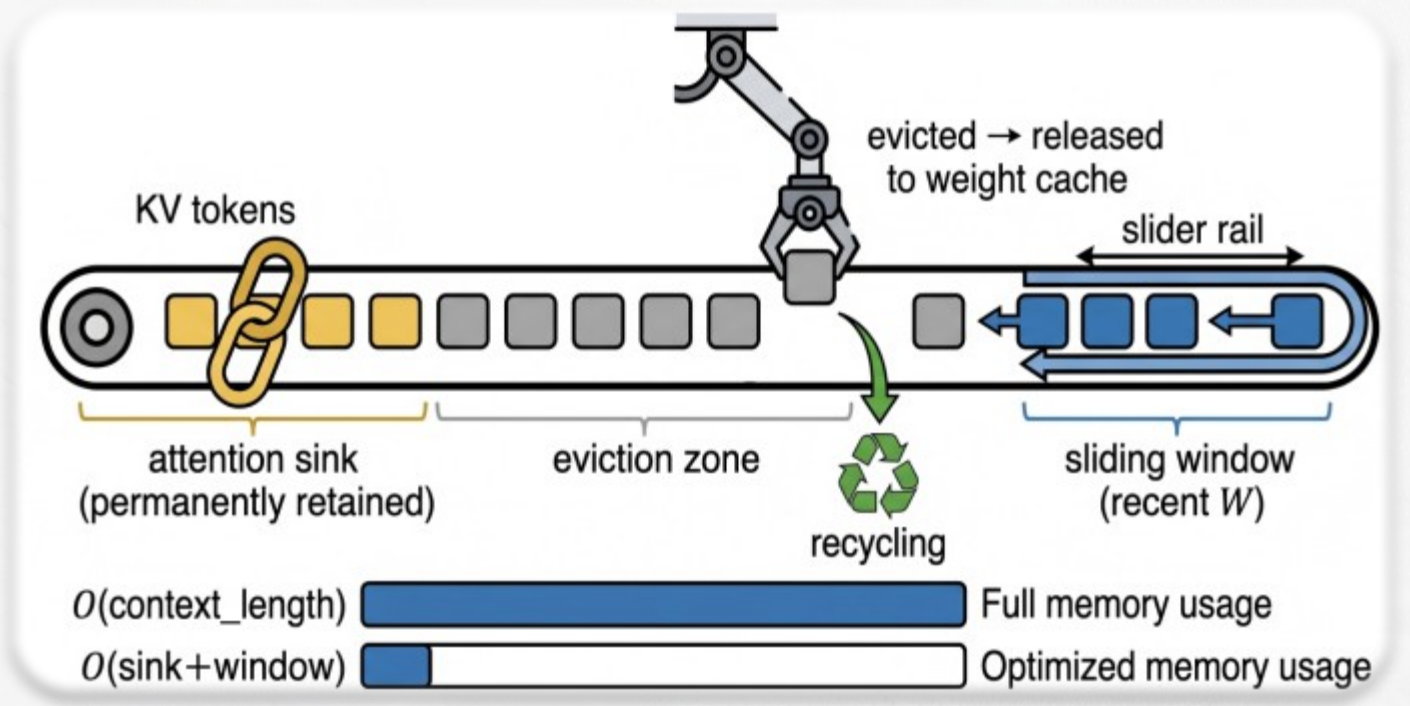
- 
- “ ”
- 45GB 8GB OOM
- 
- posix\_madvise(MADV\_RANDOM)
- Page
- Fault





# StreamingLLM

- 
- KV Cache
- OOM
- “ ”
- **StreamingLLM** 4 Token
- Attention Sink)
- 
- Token
- KV



4

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SLIM- RC

|

- SLIM\_ARC\_DISABLE=1 upstream baseline
- SLIM\_ARC\_NO\_MADV\_RANDOM=1  
MADV\_RANDOM
- SLIM\_ARC\_NO\_PREFETCH=1
- SLIM\_ARC\_DYNAMIC\_MADV=1 prefill/decode

| llama.cpp 4

- llama-model-loader.cpp +65 MADV\_RANDOM
- llama-context.cpp +85 graph\_compute ID
- llama-kv-cache.cpp +7 clear() KV
- CMakeLists.txt +4

| SLIM-ARC

|                   |          |
|-------------------|----------|
|                   |          |
| prefetch          |          |
| unified-scheduler | I/O      |
| kv-eviction       | KV Cache |
| on-demand         | pread    |

```
1  madv_block = ""mmaps_used.emplace_back(mapping->size(), 0);
2
3      // SLIM-ARC: MADV_RANDOM for large models (>6GB) to enable demand paging.
4      // Tradeoff: prefill slower (no readahead), decode 3-4x faster (MoE sparsity).
5      // SLIM_ARC_NO_MADV_RANDOM=1 to disable.
6      {
7          bool slim_arc_disabled = getenv("SLIM_ARC_DISABLE") != nullptr;
8          bool no_madv = getenv("SLIM_ARC_NO_MADV_RANDOM") != nullptr;
9          size_t msz = mapping->size();
10         if (!slim_arc_disabled && !no_madv && msz > (6ULL << 30) &&
11             mapping->addr() && msz > 0) {
12             (void) posix_madvise(mapping->addr(), msz, POSIX_MADV_RANDOM);
13             // SLIM-ARC: register mmap region for dynamic MADV switching
14             slim_arc::register_mmap_region(mapping->addr(), msz);
15         }
16     }
```



➤ >6GB



**cgroup**

/sys/fs/cgroup/memory.max

<60%



SLIM\_ARC\_DISABLE /

SLIM\_ARC\_NO\_MADV\_RANDOM

M

# ◆ MoE Router Hook



```
1 void prefetch_scheduler::worker_loop() {
2     while (true) {
3         int target_layer;
4         uint64_t sig;
5         {
6             std::unique_lock<std::mutex> lk(mtx_);
7             cv_.wait(lk, [this] { return stop_ || target_layer_ != current_layer_.load(); });
8         };
9         if (stop_) return;
10        target_layer = target_layer_;
11        sig = target_signature_;
12    }
13    if (sig != signature_.load()) continue; // stale
14
15    current_layer_.store(target_layer);
16
17    // Prefetch layers [target+1, target+window]
18    size_t bytes_this_round = 0;
19    for (int w = 1; w <= window_; ++w) {
20        int layer = target_layer + w;
21        if (layer < 0 || (size_t)layer >= tensors_by_layer_.size()) continue;
22        for (const auto & t : tensors_by_layer_[layer]) {
23            if (t.addr == nullptr || t.size == 0) continue;
24            // posix_madvise is thread-safe and idempotent
25            (void) posix_madvise(t.addr, t.size, POSIX_MADV_WILLNEED);
26            bytes_this_round += t.size;
27        }
28    }
29
30    total_bytes_.fetch_add(bytes_this_round);
31    total_calls_.fetch_add(1);
32 }
33 }
```

```
1 if (status == GGML_STATUS_SUCCESS) {
2     if (auto * s = slim_arc::get_global_prefetch_scheduler()) {
3         int n_nodes = ggml_graph_n_nodes(gf);
4         for (int i = 0; i < n_nodes; ++i) {
5             struct ggml_tensor * t = ggml_graph_node(gf, i);
6             if (!t || !t->name) continue;
7             if (strstr(t->name, "ffn_moe_topk") != nullptr && t->data != nullptr) {
8                 int layer = slim_arc::tensor_layer_from_name(t->name);
9                 if (layer < 0) continue;
10                int n_expert_used = (int) t->ne[0];
11                if (n_expert_used <= 0) continue;
12                const int32_t * ed = (const int32_t *) t->data;
13                std::vector<int> ue;
14                for (int e = 0; e < n_expert_used && e < 64; ++e) {
15                    int eid = (int) ed[e];
16                    if (eid >= 0) {
17                        bool found = false;
18                        for (int x : ue) if (x == eid) { found = true; break; }
19                        if (!found) ue.push_back(eid);
20                    }
21                }
22                if (!ue.empty()) s->cache_router_experts(layer, ue.data(), (int) ue.size());
23                break;
24            }
25        }
26    }
27 }
```



```

1 static constexpr double WEIGHT_RATIOS[5][3] =
2 { // {weight, kv, expert}
3   {0.60, 0.10, 0.30}, // PREFILL_SHORT
4   {0.50, 0.20, 0.30}, // PREFILL_LONG
5   {0.70, 0.20, 0.10}, // DECODE_SHORT
6   {0.30, 0.60, 0.10}, // DECODE_LONG
7   {0.20, 0.20, 0.60}, // MOE_DECODE
8 };

```



KV

MoE

I/O



5

× 3

I/O = 15



```

1 if (current_stats_.weight_stalls > 0) {
2   // Weight prefetch is stalling compute → increase weight budget
3   double adjustment = std::min(0.1, current_stats_.weight_stalls * 0.02);
4   budget.weight_bytes = (size_t)(budget.weight_bytes * (1.0 + adjustment));
5   budget.expert_bytes = (size_t)(budget.expert_bytes * (1.0 - adjustment * 0.
6 5)); budget.kv_bytes = (size_t)(budget.kv_bytes * (1.0 - adjustment * 0.5));
7 }

```

# StreamingLLM KV Eviction



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```
1 kv_evict_block = ""
2 // SLIM-ARC Phase 4: StreamingLLM-style KV eviction after decode.
3 // Keeps first N tokens (attention sinks) + last W tokens (sliding window),
4 // evicts tokens in between via llama_memory_seq_rm to cap KV memory.
5 // Enabled via SLIM_ARC_KV_EVICT=1, configured by SLIM_ARC_KV_SINK/WINDOW.
6 if (status == GGML_STATUS_SUCCESS && !batched) {
7     static bool kv_evict_enabled = getenv("SLIM_ARC_KV_EVICT") != nullptr;
8     if (kv_evict_enabled) {
9         static int kv_sink = []{ const char * e = getenv("SLIM_ARC_KV_SINK"); return e ? atoi(e) : 4; }();
10        static int kv_window = []{ const char * e = getenv("SLIM_ARC_KV_WINDOW"); return e ? atoi(e) : 1024; }();
11        static int last_evicted_pos = 0;
12        static bool first_call = true;
13        if (first_call) {
14            fprintf(stderr, "SLIM-ARC KV eviction: ENABLED (sink=%d, window=%d)\n", kv_sink, kv_window);
15            first_call = false;
16        }
17        llama_pos pos_max = memory->seq_pos_max(0);
18        int seq_len = (int) pos_max + 1;
19        int threshold = kv_sink + kv_window;
20        if (seq_len > threshold) {
21            int p0 = (last_evicted_pos > kv_sink) ? last_evicted_pos : kv_sink;
22            int p1 = seq_len - kv_window;
23            if (p1 > p0) {
24                memory->seq_rm(0, p0, p1);
25                last_evicted_pos = p1;
26                fprintf(stderr, "SLIM-ARC KV eviction: evicted [%d,%d] (seq_len %d->%d)\n",
27                    p0, p1, seq_len, seq_len - (p1 - p0));
28            }
29        }
30    }
31 }
```

➤ sink token

✦

KV

decode ✦9.6%

KV

$O(L)$

$O(1)$

➤ KV Cache clear()

posix\_madvise(DONTNEED)

scripts/apply-

slim-arc.py:345-356

5

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SLIM- RC

7

| CPU | i9-13900H (6P+8E, 20T) |
|-----|------------------------|
| RAM | 32 GB DDR5 (WSL2)      |
|     | NVMe SSD (~3.5 GB/s)   |
|     | AVX2 + AVX_VNNI + FMA  |
|     | llama.cpp + SLIM-ARC   |

8

| Qwen3-4B       | Q4_K_M | 2.4 GB  | —   |
|----------------|--------|---------|-----|
| OLMoE-1B-7B    | Q4_K_M | 3.9 GB  | 64  |
| Qwen3-Next-80B | IQ4_XS | 39.7 GB | 512 |

1

- drop\_caches
- page cache
- llama-bench pp/tg
- 3
- GSM8K 8-shot + WikiText-2
- PPL LEM
- upstream llama.cpp
- mmap+WILLNEED+KV f16
- SLIM- RC

| IQ4\_XS

| ( )        | pp   | tg       |  |
|------------|------|----------|--|
| 8 GB (4 )  | —    | —        |  |
| 16 GB (8 ) | —    | 2.27 t/s |  |
| 32 GB (8 ) | 4.44 | 3.03 t/s |  |

| Q4\_K\_M

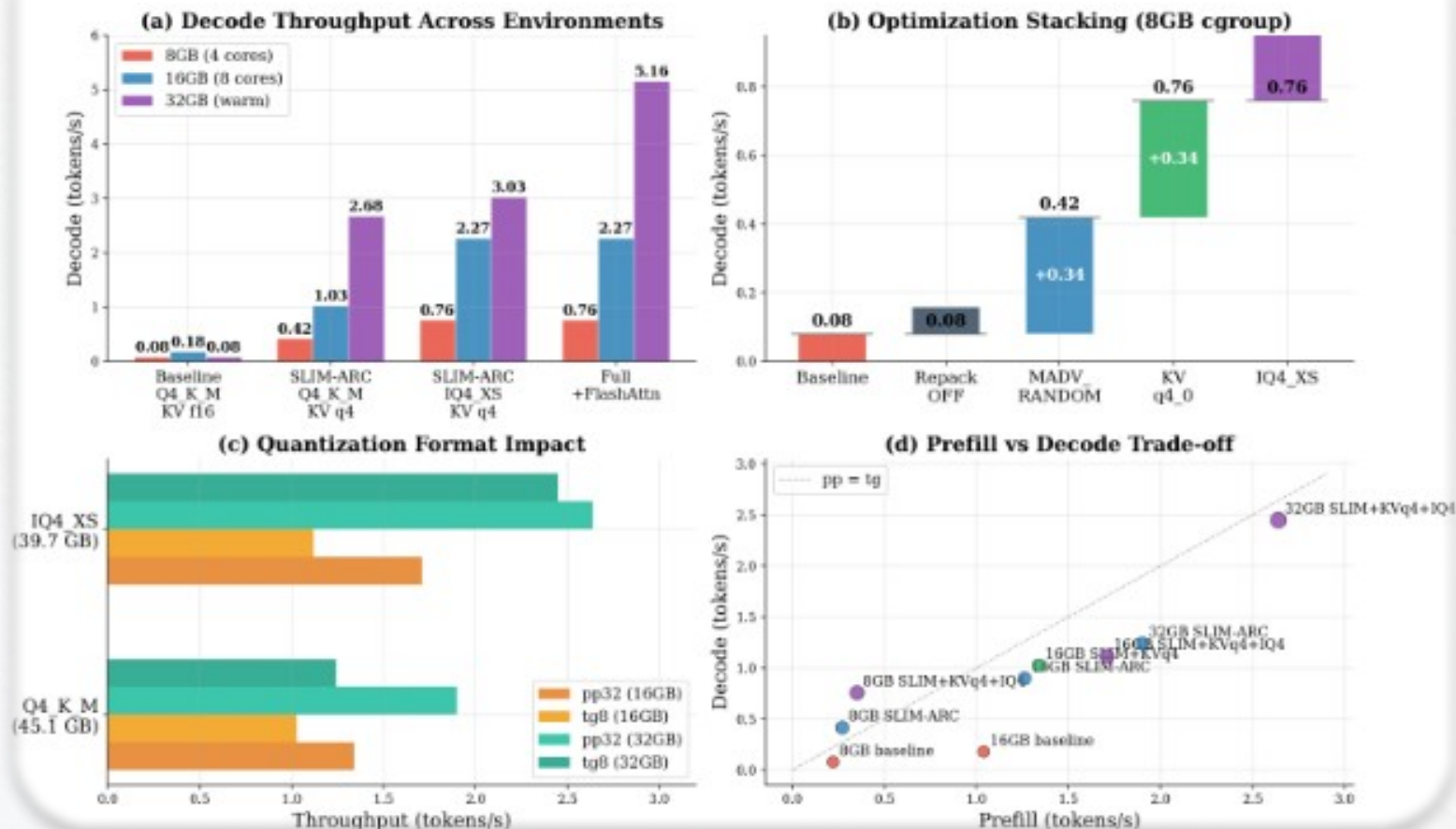
| ( )        | pp   | tg       |               |
|------------|------|----------|---------------|
| 8 GB (4 )  | —    | —        |               |
| 16 GB (8 ) | 1.96 | —        | tg            |
| 32 GB (8 ) | —    | 2.68 t/s | IQ4_XS<br>12% |

- 8GB 40–45GB page fault  
0.76 t/s 9.5×
- 16GB 2.27 t/s 80B 6.2×
- 32GB 3.03 t/s IQ4\_XS
- Q4\_K\_M 13%

—— SLIM- RC



## SLIM-ARC Performance Landscape: Qwen3-Next-80B



- **64.5×** 8GB baseline 0.08 → 32GB 5.16
- **32GB** Flash attention **+71.4%** 3.01→5.16
- 

8GB→32GB  
SLIM-ARC

| 2GB+1

|             |        | tg16 |  |
|-------------|--------|------|--|
| Qwen3-4B    | 2.4 GB | 2.58 |  |
| OLMoE-1B-7B | 3.9 GB | 8.92 |  |

➤ OLMoE MoE 2GB+1  
8.92 t/s MADV\_RANDOM

➤ IQ4\_XS Q4\_K\_M 0.25 bit  
GSM8K 50% 0%  
" "

➤ PPL MoE GSM8K

| GSM8K

|          |             | GSM8K | t/s |
|----------|-------------|-------|-----|
| Qwen3-4B | Q4_K_M      | 75%   | 7.8 |
| 80B      | Q4_K_M+KVq4 | 50%   | 0.7 |
| 80B      | IQ4_XS+KVq4 | 0%    | 1.7 |

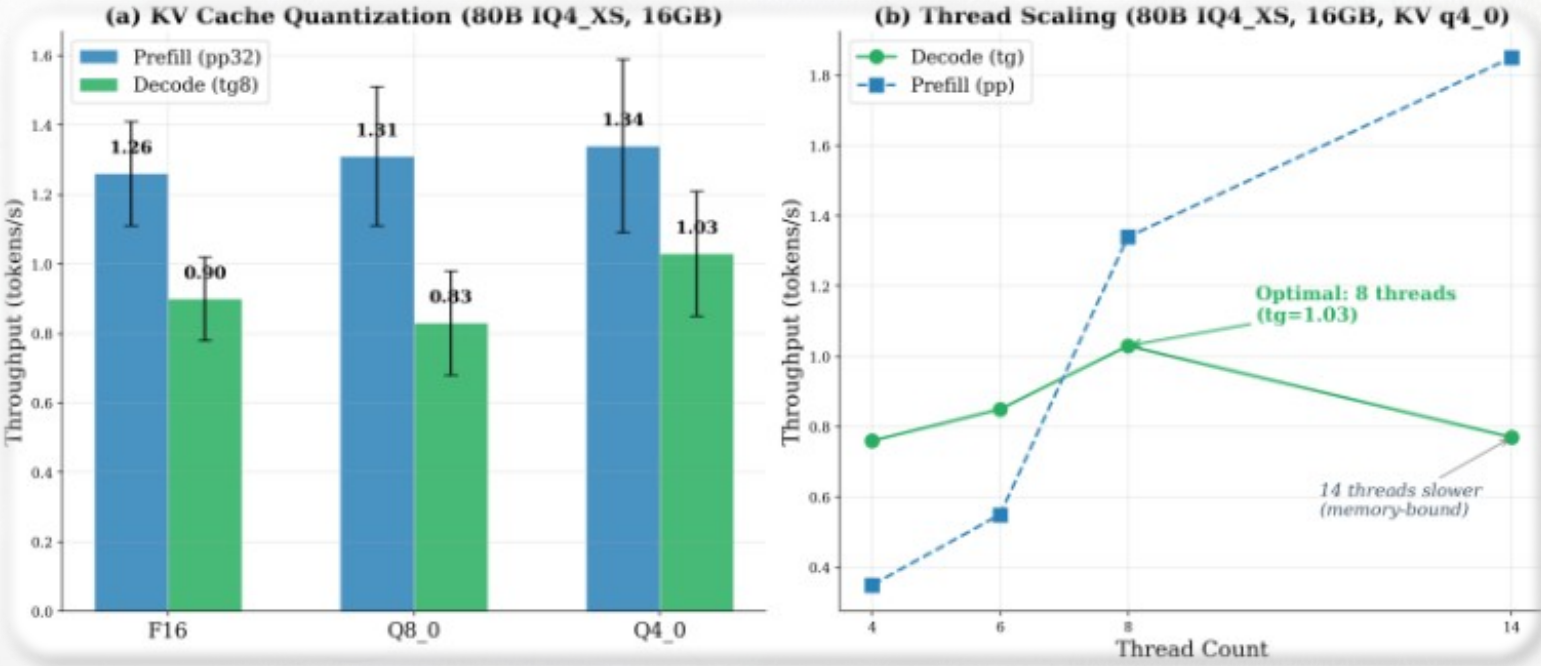
# KV Cache

| KV

| KV   | pp64 | tg48 | KV   | vs Q4_0 |
|------|------|------|------|---------|
| F16  | —    | 3.92 | 2×   | ↗4%     |
| Q8_0 | 5.37 | 2.91 | 1×   | -23%    |
| Q4_0 | 4.44 | 3.03 | 0.5× | best    |

KV □□□□

- Q4\_0 KV 6GB→3GB
- 32GB F16 ↗4%
- 8GB Q4\_0 +14%
- 



- 8
- 
- decode memory-bound NVMe

|                  | pp64 | tg48 | vs Full |    |
|------------------|------|------|---------|----|
| Full SLIM-ARC    | 4.44 | 3.03 | —       |    |
| - KV q4_0 (f16)  | —    | 3.92 | +4%     |    |
| - MADV (DISABLE) | 3.72 | 2.15 | -41%    |    |
| +Eviction        | —    | 3.30 | -12%    | FA |

□ □

➤ MADV\_RANDOM decode

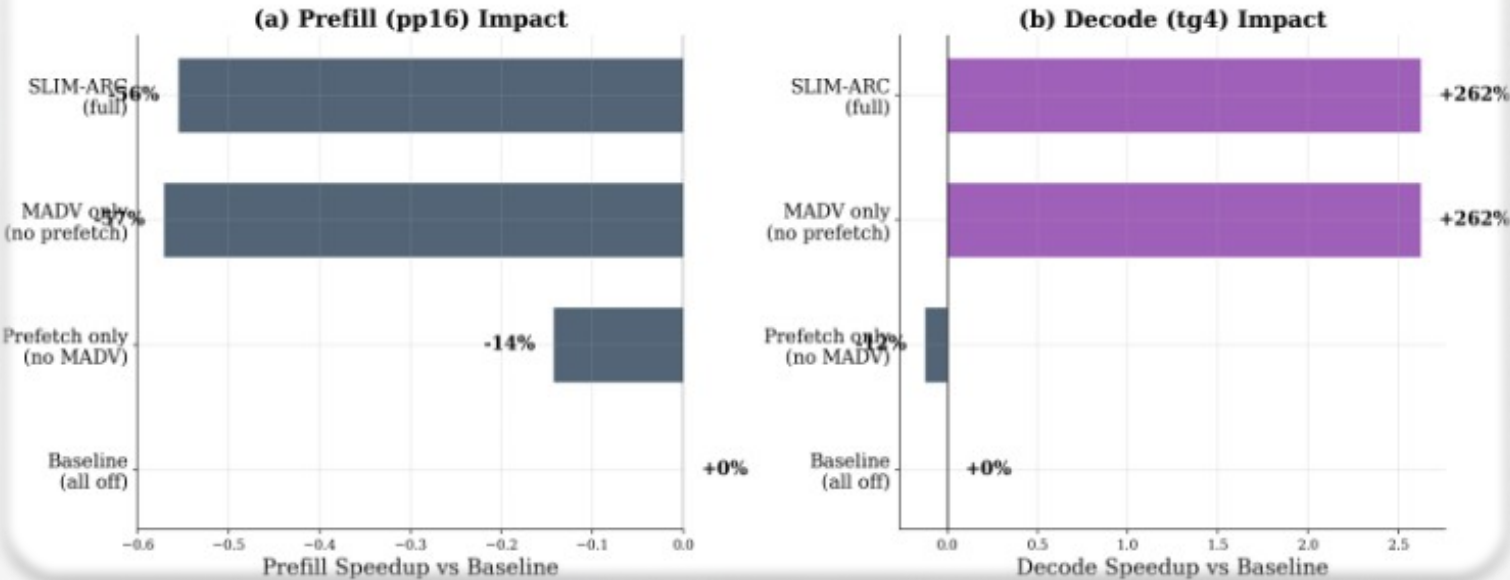
- 41%

➤ Qwen3-Next-80B

token 512 10

2%

Four-Way Ablation: Component Contribution Analysis (80B Q4\_K\_M, 8GB)

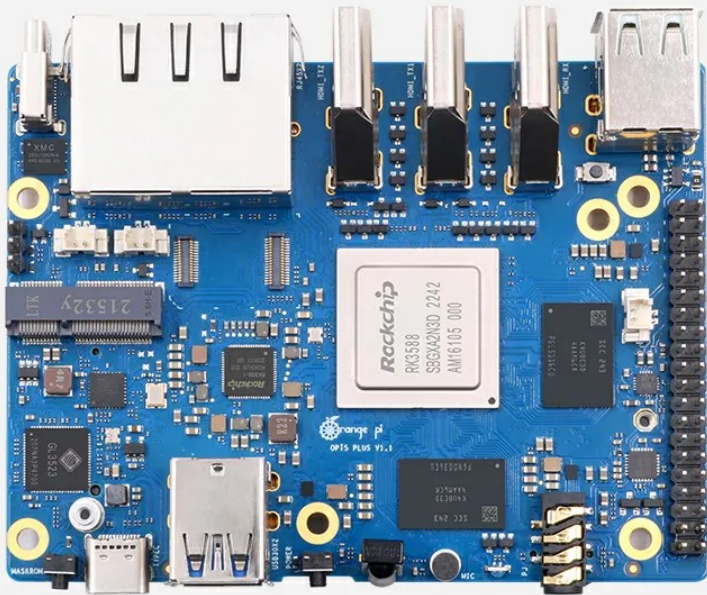


➤ MADV\_RANDOM ⇒ KV q4\_0

➤ Eviction VS Flash Attention

Orange Pi 5 Plus

- 瑞芯微RK3588八核 64位处理器
- 主频高达2.4GHz
- 4GB/8GB/16GB ( LPDDR4/4x )
- 支持8K视频解码
- 可以外接16GB/32GB/64GB/128GB/256GB eMMC模块
- 2.5G高速双网口
- 支持2230 Wi-Fi 6/BT模块的M.2插槽
- 支持2280 SSD的M.2插槽
- 2个USB 3.0 , 2个USB 2.0



|   |               |                |            |
|---|---------------|----------------|------------|
| ➤ | 45GB          | Qwen3-Next-80B | 8GB RK3588 |
|   |               | RSS            | OOM        |
| ➤ |               | ~6.3GB         |            |
|   |               | 4.5%→34.9%     |            |
|   |               | 31%→55%        | -56%       |
| ➤ | -c 8192~16384 |                | OOM RSS    |
|   | ~6.5GB        | KV eviction    |            |



6

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SLIM- RC

# Live Demo



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localhost:8090

SLIM-ARC Live Demo  
Qwen3-Next-80B-IQ4\_XS · 38 GB · 32GB (mmap on-demand)

Qwen3-Next-80B-IQ4\_XS (38 GB) 32GB (mmap on-demand)

与 SLIM-ARC 优化后的模型对话，体验端侧 MoE 大模型推理

Assistant

你好！我是 SLIM-ARC 优化后的端侧大模型。在下方输入问题，我会流式回答。右侧面板展示实时系统状态。

User

你是谁

Assistant

我是基于 80B MoE的端侧AI助手，受MADV\_RANDOM优化，在8GB内存下运行，能帮你解答问题，无需云端。

User

moe是什么

Assistant

M是指"Mixture of Experts，即"混合专家模型。它通过激活少量参数，实现高效推理，比传统大模型更节省资源，同时保持高性能，是 SLIM-ARC优化

输入问题，回发送...

发送

系统内存

已用 / 总计 5.53 GB / 33.66 GB

Cache (文件缓存) 27.92 GB

推理速度

5.13 t/s

已生成 token 42

MOE 专家激活

10 / 512 激活 稀疏率 98.0%

SLIM-ARC 优化链

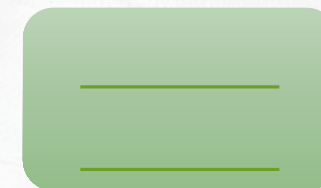
A 内核协同 MADV\_RANDOM 按需加载

A 内核协同 REPACK=OFF (禁用翻倍)

B 量化压缩 KV q4\_0 量化

B 量化压缩 Qwen3-Next-80B-IQ4\_XS (38 GB)

C 计算融合 FlashAttention



|   |                        |                |              |                |                  |        |  |   |
|---|------------------------|----------------|--------------|----------------|------------------|--------|--|---|
| ➤ | M DV_R NDOM MoE decode |                |              |                |                  |        |  |   |
|   | decode                 | —              | tg48         | 3.03           | 2.15 t/s         | -29%   |  |   |
| ➤ | GGML_CPU_REP CK        |                |              |                |                  |        |  |   |
|   | repack                 | -DGGML_CPU_REP | CK=OFF       | CPU backend    |                  | mmap   |  |   |
| ➤ |                        |                |              |                |                  |        |  |   |
|   | → C                    | '              | M DV         | B              | KV q4_0 / IQ4_XS | '      |  | B |
| ➤ |                        |                |              |                |                  |        |  |   |
|   | 80B MoE                |                |              |                |                  |        |  |   |
|   | 32 GB                  |                | IQ4_XS       | Qwen3-Next-80B | 5.16 tokens/s    | decode |  |   |
|   | Flash                  | ttention       |              |                |                  |        |  |   |
| ➤ |                        |                |              |                |                  |        |  |   |
|   |                        |                | RK3588 8GB / | 5 4GB          |                  |        |  |   |
|   | RK3588                 | 80B            | OOM          | RSS 6.3GB      | 31→55%           | 5      |  |   |
|   |                        | Qwen3-4B       | OOM          | RSS 2.42GiB    | •6GB             |        |  |   |

# LIM-ARC

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Website: <https://slim.nexa-lang.com/>

<https://www.bilibili.com/video/BV1fXTF6HEAw?p=2>

|  |                             |
|--|-----------------------------|
|  |                             |
|  |                             |
|  | pressure × waste            |
|  | model-owned runtime ♦ lease |
|  | schema-1 ♦                  |
|  |                             |

|       |  |  |      |
|-------|--|--|------|
|       |  |  |      |
| 64.5× |  |  |      |
| 850%  |  |  |      |
|       |  |  | JSON |

- 
- /
  - TPS
  - promotion results.json finals-
  - run ID raw
  - manifest
  - cold / warm

## generation-token

|  |                       |
|--|-----------------------|
|  |                       |
|  | prefetched - selected |
|  | expert slice          |
|  | POSIX_MADV_DONTNEED   |
|  |                       |

|   |          |  |  |
|---|----------|--|--|
|   | selected |  |  |
|   |          |  |  |
|   |          |  |  |
| / |          |  |  |

- prefetch generation token
- settle / cancel generation
- settle
- flag-off DONTNEED
-

## | pressure

|          | /                  |
|----------|--------------------|
| normal   | 85%                |
| high     | ≥ 85%              |
| critical | ≥ 95%              |
| missing  | /                  |
|          | ≤90% / ≤75%      2 |

| pressure | stable | temporal | hot |
|----------|--------|----------|-----|
| normal   | *      | *        | *   |
| high     | *      |          |     |
| critical |        |          |     |

## | opt-in

- cgroup v2 pressure snapshot
- invalid / overflow missing fallback
- waste EWM ≥60% stable experts
- 2 /
- flag-off

| model-owned runtime ♦ lease

| owner      | llama_model::impl          |
|------------|----------------------------|
| activate   | mmap                       |
| acquire    | move-only runtime lease    |
| deactivate | →→→ join                   |
| destroy    | runtime      model mapping |

|

| 2 GiB  | pp64/tg16 | exact 2 | base=0       |
|--------|-----------|---------|--------------|
| 4 CPU  | same seed | cold 1  | patch=1      |
| swap 0 | 5 configs | warm 1  | schema<br>-1 |

|

- (generation, layer)
- scheduler mutex      madvise
- 
- 
- cold / warm

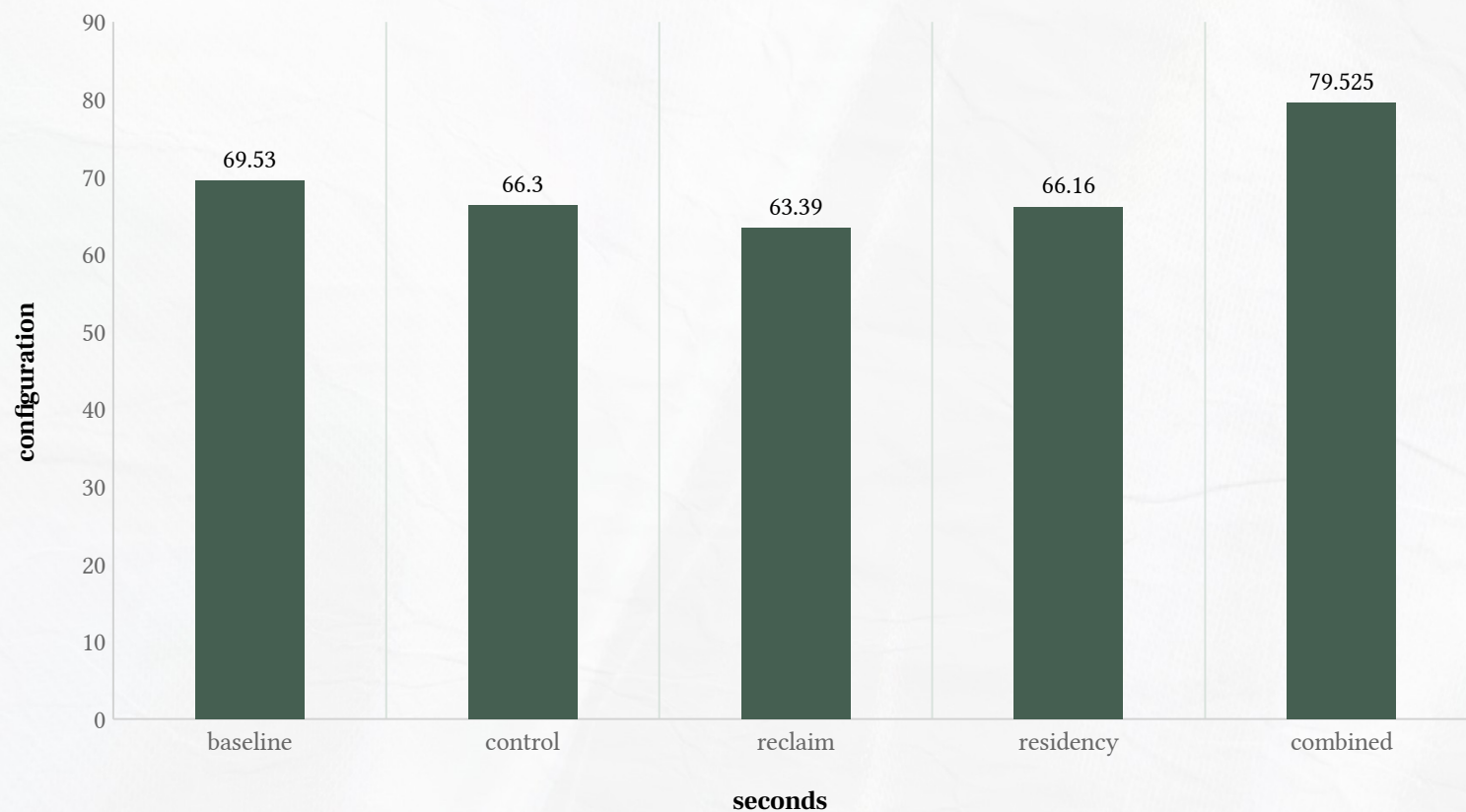
2 GiB

cold



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SUN YAT-SEN UNIVERSITY

### Cold wall time CCCCCC



/ / cold-warm  
20/20  
➤ control cold 66.300 s  
➤ combined cold 79.525 s  
➤ control 19.95%  
➤ memory.peak  
2.000 GiB

combined =  
rejected

| cache-separated

|           | cold / warm / overall           |
|-----------|---------------------------------|
| reclaim   | kept / kept / kept opt-in       |
| residency | kept / kept / kept opt-in       |
| combined  | rejected / kept / rejected      |
|           | 2 rounds × 5 configs × 2 caches |
|           | 20/20 success    peak=2.000 GiB |

|

|           | runs |                   |                 |
|-----------|------|-------------------|-----------------|
| reclaim   | 8    | 0 calls           |                 |
| residency | 8    | 111.44 GB<br>skip | fail-<br>closed |
| combined  | 4    | +19.95%<br>cold   |                 |

|

- reclaim      8      /
- residency    6144      critical  
sample                    111.44 GB
- 2 GiB      admitted=0  
fallback=0                fail-closed
- source / patched tree /  
image / model / cgroup
- pre-linkage                64.5× / 850%